Deep Segmentation-Emendation Model for Gland Instance Segmentation

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Abstract. Accurate and automated gland instance segmentation on histology microscopy images can assist pathologists to diagnose the malignancy degree of colorectal adenocarcinoma. To address this problem, many deep convolutional neural network (DCNN) based methods have been proposed, most of which aim to generate better segmentation by improving the model structure and loss function. Few of them, however, focus on further emendating the inferred predictions, thus missing a chance to refine the obtained segmentation results. In this paper, we propose the deep segmentation-emendation (DSE) model for gland instance segmentation. This model consists of a segmentation network (Seg-Net) and an emendation network (Eme-Net). The Seg-Net is dedicated to generating segmentation results, and the Eme-Net learns to predict the inconsistency between the ground truth and the segmentation results generated by Seg-Net. The predictions made by Eme-Net can in turn be used to refine the segmentation result. We evaluated our DSE model against five recent deep learning models on the 2015 MICCAI Gland Segmentation challenge (GlaS) dataset and against two deep learning models on the colorectal adenocarcinoma (CRAG) dataset. Our results indicate that using Eme-Net results in significant improvement in segmentation accuracy, and the proposed DSE model is able to substantially outperform all the rest models in gland instance segmentation on both datasets.

1 Introduction

Colorectal adenocarcinoma is one of the most common cancers in the world [1]. Accurate segmentation of the gland instances on histology images is an effective means to assess the gland morphology, which is essential for pathologists to determine the malignancy degree of colorectal adenocarcinoma [2,3]. Manual annotation of gland instances, however, requires a high degree of concentration and

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expertise and is time-consuming due to the myriad size of histology images [4]. Therefore, automated gland instance segmentation is of crucial importance in clinical practice to improve the efficiency as well as objectivity and to reduce the workload of pathologists. The automation of this task is challenging since (1) the heterogeneous glandular morphology between benign and malignant makes it difficult to delineate the glands from the background, and (2) the tiny gaps between adjacent glands render more difficulties in separating each gland from others. Two examples that illustrate both issues are displayed in Fig. 1.

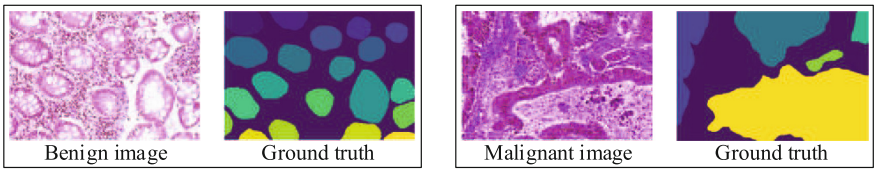

Fig. 1. Two examples of gland instance segmentation in benign and malignant cases. Individual glands are denoted by different colors. (Color figure online)

Recently, deep convolutional neural networks (DCNNs) have achieved remarkable success in image segmentation [2,4–7], which inspires many researchers to apply DCNNs to gland instance segmentation. Chen *et al.* [5] proposed a multi-task deep neural network that employs multi-level contextual features to separate the glands from the background and also explores the contour information to delineate individual glands. Grahama *et al.* [2] presented a minimal information loss dilated network to retain maximum information during feature extraction, aiming to improve the segmentation of individual glands. Xu *et al.* [6] developed a deep multi-channel network that uses three channels to learn gland regions, contours, and location cues, respectively, to segment the glands and separate individual ones. Yan *et al.* [4] proposed a shape-preserving loss to learn gland segmentation and contour detection simultaneously. These methods aim to generate better predictions through improving the segmentation model, such as designing advanced network architectures, defining novel losses, and supplementing additional contour detection modules. Different from these attempts, we focus on refining the inferred segmentation results (i.e. emendating the under-segmentation or over-segmentation regions), which is beneficial to further improve the segmentation of gland instances.

In this paper, we propose a deep segmentation-emendation (DSE) model for the gland instance segmentation on histology images. This model is composed of two networks: a segmentation network (Seg-Net) and an emendation network (Eme-Net). The Seg-Net predicts the segmentation mask (MaskSeg) of an input image, and the Eme-Net learns an emendation mask (MaskEme) that represents the inconsistency between the predicted MaskSeg and the corresponding ground truth. In the inference stage, each test image is fed to both trained Seg-Net and Eme-Net, and the predicted MaskEme is used to emendate the predicted

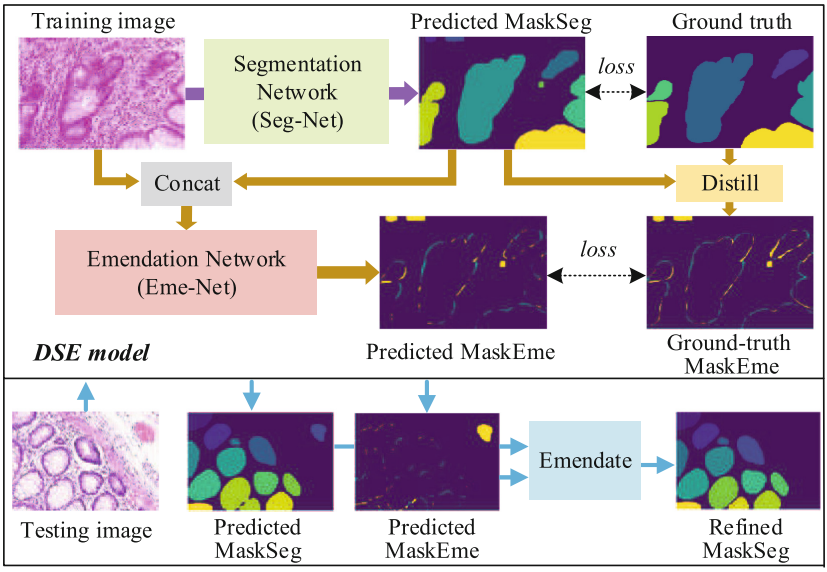

Fig. 2. The technical pipeline of our proposed DSE model.

MaskSeg for an improved segmentation performance. We evaluated the proposed DSE model against several existing approaches on two benchmark datasets.

The contributions of this work include: (1) supplemented the Seg-Net with the Eme-Net, which is able to predict the over- and under-segmentation resulted from the Seg-Net; (2) employed such predictions to emendate the segmentation results produced by the Seg-Net; and (3) reported the state-of-the-art gland instance segmentation performance on two benchmark datasets.

2 Method

The pipeline of our DSE model consists of offline training of both Seg-Net and Eme-Net and online inference, in which the predicted MaskEme is employed to emendate the MaskSeg predicted by Seg-Net for refinement (see Fig. 2).

Segmentation Network: We employ the advanced semantic segmentation network Deeplabv3+ [8] pre-trained on MS-COCO [9] and PASCAL VOC 2012 datasets [10], as the backbone of Seg-Net. To adapt Deeplabv3+ to our segmentation task, we remove its last convolutional layer, and then add a new convolutional layer with two output channels and a softmax layer for prediction. The weights attached to new layers are randomly initialized. We optimize the Seg-Net by minimizing the cross-entropy loss.

Emendation Network: To further improve the quality of the inferred MaskSeg, we design the Eme-Net that learns a MaskEme to estimate the segmentation

quality of Seg-Net and emendate the wrong predictions. We define the ground-truth MaskEme as the pixel-wise consistency between the binary MaskSeg predicted by Seg-Net and its ground truth. Formally, the value of the i-th pixel in the ground-truth MaskEme, $\hat{E}_i$, can be denoted by:

$$\hat{E}_i = \begin{cases} 0 & if\ S_i = \hat{S}_i \\ 1 & if\ S_i < \hat{S}_i \\ 2 & if\ S_i > \hat{S}_i \end{cases} \tag{1}$$

where S_i and $\hat{S}_i$ represent the value of the i-th pixel in the predicted MaskSeg and segmentation ground truth, respectively. If $\hat{E}_i = 0$, it means that the prediction S_i is accurate and no emendation is needed. Otherwise, the prediction is either under-segmentation (i.e. $\hat{E}_i = 1$) or over-segmentation (i.e. $\hat{E}_i = 2$), and need be emendated in the inference stage.

We use the trained Seg-Net as the backbone of Eme-Net for a better initialization. The original image and the predicted MaskSeg are concatenated as the input to Eme-Net. The 4-th input channel (i.e., predicted MaskSeg) is initialized by the average weights of other three channels (i.e., RGB image). The output channel of the last convolutional layer is changed to three, and the weights of the modified convolutional layer are randomly initialized.

Different from training Seg-Net, training Eme-Net suffers from extremely severe class-imbalance, since the number of pixels with a value of 1 or 2 accounts for only about 1% of the pixels with a value of 0 in each ground-truth MaskEme. To address this issue, we jointly optimize the weighted multi-class Dice loss and the cross-entropy (ce) loss during the process of training Eme-Net. The combined loss can be expressed as:

$$\mathcal{L} = \mathcal{L}_{Dice} + \mathcal{L}_{ce} = \frac{1}{C}\sum_{c=1}^{C}\omega_c\left[1 - \frac{2\sum_{i=1}^{V}\hat{E}_i^c E_i^c}{\sum_{i=1}^{V}(\hat{E}_i^c + E_i^c) + \varepsilon}\right] - \frac{1}{V}\sum_{v=1}^{V}\sum_{c=1}^{C}\omega_c \hat{E}_i^c log E_i^c \tag{2}$$

where C is the number of categories, V denotes the number of pixels, E_i^c indicates the prediction of i-th pixel belonging to the class c, $\hat{E}_i^c$ represents the target value of i-th pixel in the MaskEme, ε is a smoothing factor, and ω_c is the weight of the class c. The weight ω_c can be formulated as $\omega_c = log\frac{V}{V^c}$, where V^c denotes the number of pixels belonging to the class c.

Inference Process: The predicted MaskEme can be used to emendate the wrong predictions (i.e. the over- and under-segmentation regions) inferred by the Seg-Net in inference stage for refinement. The value of i-th pixel for the refined MaskSeg $S_i^{'}$ can be formulated as:

$$S_i^{'} = \begin{cases} S_i & if\ E_i = 0 \\ 1 & if\ E_i = 1 \\ 0 & if\ E_i = 2 \end{cases} \tag{3}$$

where S_i and E_i represent the value of i-th pixel of the predicted MaskSeg and MaskEme, respectively. If $E_i = 1$, we emendate the prediction S_i as the gland class $\{1\}$. If $E_i = 2$, we emendate the prediction S_i as the background class $\{0\}$.

3 Datasets

We evaluated the proposed DSE model on two benchmark datasets: the 2015 MICCAI Gland Segmentation (GlaS) Challenge dataset [11] and the colorectal adenocarcinoma gland (CRAG) dataset [2,12]. The GlaS dataset contains 85 training and 80 test images (60 in Part A and 20 in Part B). The image size ranges from 567×430 to 775×522. The CRAG dataset has 173 training and 40 test images, mostly having a size of 1512×1516. All training images in both datasets are equipped with the ground-truth masks for instance segmentation, which were manually annotated by pathologists.

4 Experiments and Results

Implementation Details: During training, we randomly cropped patches from each training image as the input of both Seg-Net and Eme-Net to not only reserve as much information as possible but also weigh the balance of segmentation performance and computational cost. The patch size was set to 416×416 on the GlaS dataset and 512×512 on the CRAG dataset. To alleviate the overfitting of deep neural networks, we employed the online data argumentation techniques, including random rotation, shear, shift, zoom of width and height, whitening, horizontal and vertical flips, and color normalization, to enlarge the training set. We optimized both networks using the standard SGD with a batch size of 4, an initial learning rate of 0.001, and the maximum epoch of 1000, and also set 20% of the training set aside to monitor the performance of both networks. In the inference stage, we extracted patches of the same size from each test image with a stride of 256 pixels. The final segmentation of a whole image is generated by recomposing and averaging the predictions of those patches.

Evaluation Metrics: We evaluated the segmentation results using three metrics provided by the GlaS Challenge[1], including (1) accuracy of the detection of individual glands (Object F1 Score), (2) volume-based accuracy of the segmentation of individual glands (Object Dice), and (3) shape similarity between the segmentation result and its ground truth (Object Hausdorff Distance). For the overall results, each method will be assigned three ranking numbers based on the three metrics. The sum of these numbers will be used for the final ranking.

Results on the GlaS Challenge Dataset: We compared the proposed DSE model against the minimal information loss dilated network (MILD-Net) [2], the deep network with shape-preserving loss (SPL-Net) [4], the deep multi-channel network (DMCN) [6], the deep contour-aware network (DCAN) [5], and the

[1] https://warwick.ac.uk/fac/sci/dcs/research/tia/glascontest/.

Table 1. Comparison with five methods on the GlaS Challenge dataset. S and R denote score and rank, respectively

Methods	Object F1				Object Dice				Object Hausdorff				Rank sum
	Part A		Part B		Part A		Part B		Part A		Part B		
	S	R	S	R	S	R	S	R	S	R	S	R	
DSE model (Ours)	**0.926**	**1**	**0.862**	**1**	**0.927**	**1**	**0.871**	**1**	**31.209**	**1**	**80.509**	**1**	**6**
MILD-Net [2]	0.914	3	0.844	2	0.913	2	0.836	3	41.540	2	105.890	2	14
SPL-Net [4]	0.924	2	0.844	2	0.902	4	0.840	2	49.881	5	106.075	3	18
DMCN [6]	0.893	5	0.843	4	0.908	3	0.833	4	44.129	3	116.821	4	23
DCAN [5]	0.912	4	0.716	5	0.897	5	0.781	6	45.418	4	160.347	6	30
MPCNN [11]	0.891	6	0.703	6	0.882	6	0.786	5	57.413	6	145.575	5	34

Table 2. Comparison with two methods on the CRAG dataset.

Methods	Object F1		Object Dice		Object Hausdorff		Rank sum
	S	R	S	R	S	R	
DSE model (Ours)	**0.835**	**1**	**0.889**	**1**	**120.127**	**1**	**3**
MILD-Net [2]	0.825	2	0.875	2	160.140	2	6
DCAN [5]	0.736	3	0.794	3	218.76	3	9

Table 3. Performance of the Seg-Net and our DSE model on two datasets.

Datasets	Method	Object F1	Object Dice	Object Hausdorff
GlaS Part A	Seg-Net	0.888	0.908	41.009
	DSE model	**0.926**	**0.927**	**31.209**
GlaS Part B	Seg-Net	0.827	0.855	89.090
	DSE model	**0.862**	**0.871**	**80.509**
CRAG	Seg-Net	0.774	0.850	155.430
	DSE model	**0.835**	**0.889**	**120.127**

multi-path convolutional neural network (MPCNN) [11] on the GlaS dataset. The obtained average performance of these models was listed in Table 1. It shows that our DSE model achieves the highest accuracy in terms of all performance metrics on both parts of testing data. Specifically, our model improves the Object F1 by 0.2%, Object Dice by 1.4%, and Object Hausdorff Distance by 10.331 over the second best performance on the Part A of testing data. The results on the Part B are generally lower than that on Part A, since Part B contains a majority of malignant images with more complex morphology that increases the difficulty of gland instance segmentation. Nevertheless, our model exhibits an even more obvious performance improvement on the Part B (i.e. 1.8% in Object F1, 3.1% in Object Dice, and 25.381 in Object Hausdorff Distance) than on Part A. It evidences that our model can improve the accuracy of gland instance segmentation and shape detection on both easy and difficult cases.

Results on the CRAG Dataset: We also evaluated the proposed DSE model against MILD-Net [2] and DCAN [5] methods on the CRAG dataset. The obtained average performance in Table 2 reveals that our DSE model consistently surpasses other two methods by large margins over all metrics, which is in line with the conclusion we drew on the GlaS dataset. The superior performance on both datasets also suggests that our model is robust to handle different datasets.

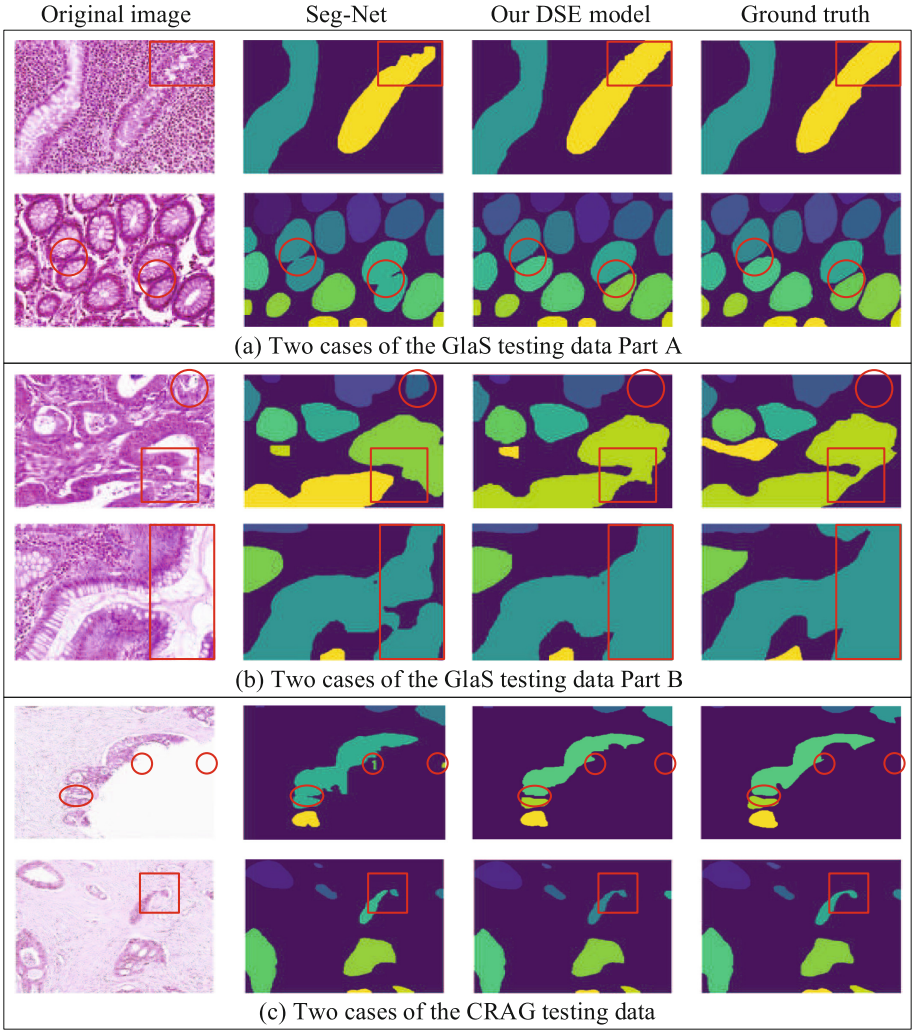

Fig. 3. Visualization of gland segmentation results of six cases selected from the GlaS and CRAG datasets. Multiple gland instances are highlighted by different colors. (Color figure online)

Ablation Study: The major uniqueness of our DSE model is to use the newly designed Eme-Net to emendate the wrong predictions made by Seg-Net. To validate the performance improvement resulted from Eme-Net, we compared our model to Seg-Net, which does not use Eme-Net for emendation, on both GlaS and CRAG datasets. The average performance of both models was shown in Table 3. It reveals that using Eme-Net to emendate the segmentation results produced by Seg-Net improves substantially the Object Dice by 1.9%, 1.6%, and 3.9% on the Part A of GlaS, Part B of GlaS, and CRAG datasets, respectively. Meanwhile, we visualized six histology images from both datasets and the corresponding segmentation results obtained by Seg-Net and our DSE model, together with the ground truth, in Fig. 3. It shows that our model can segment glands more accurately than Seg-Net, by filling the under-segmented regions (highlighted by red rectangles) and removing the over-segmented regions (highlighted by red circles). Both visual and quantitative evaluation proves the effectiveness of using Eme-Net to improve the performance of gland instance segmentation.

Computational Complexity: The DSE model was trained using the Keras software on a NVIDIA Tesla P100 GPU. In our experiments, it took about 12 h to train Seg-Net and 24 h to train Eme-Net, and cost only 0.5 to 2 s to segment each test image. Although training the model is time-consuming, it can be done offline. The fast online testing suggests that our DSE model could be used in a routine clinical workflow.

5 Conclusion

In this paper, we proposed the DSE model, a segment-and-refine paradigm, for gland instance segmentation on histology images. This model consists of two networks: Seg-Net for image segmentation and Eme-Net for result emendation. Our results on the GlaS Challenge dataset and CRAG dataset suggest that using Eme-Net to emendate the segmentation produced by Seg-Net leads to a substantial performance gain and the proposed DSE model outperforms several state-of-the-art methods by large margins on both datasets. We believe that this method is general and can be extended to the instance segmentation of other medical images, such as nuclear segmentation.

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